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A COMPREHENSIVE STUDY ON THE FACTORS SHAPING STUDENT OUTCOMES

(Group 16)

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Introduction

The purpose of this report is to investigate the determinants of students' academic performance. In particular, we investigate the extent to which playing hours, parental income, and parents' education levels determine the grades of the students. The aim of the analysis is to gain a clearer picture of the interrelationships among the variables and their influence on academic achievement. In this way, we hope to gain a deeper understanding of how these variables can account for differences in students' performance.

- **Importance of the Analysis**

It is essential to understand the role of family background and individual time management in achieving academic success, as this can translate into more effective education strategy and support mechanisms. This comparison is relevant because it informs the contribution of socioeconomic factors in determining educational attainment. Through examining the correlation between the income of parents and academic success, or the number of playing hours and academic grades, we can determine areas of improvement for intervention in student support. This can eventually guide policy in enhancing student success and equity in education.

- **Data Used**

The data for the analysis was obtained from Kaggle Datasets, which offers access to diverse datasets for analysis and research. The dataset contains students' grades, study hours, parental income, and education levels of parents. These variables have been selected as they are considered to have a major impact on academic performance and thus well-suited for this kind of analysis. The dataset is organized, making it possible for us to analyze and obtain useful information easily.

- **Objective of the Analysis**

The main objective of this analysis is to provide answers to fundamental questions regarding the contributors to students' performance. In particular, the analysis aims to provide answers to the following questions:

1. How does time spent on leisure activities (playing hours) impact the grades of students?
2. Does parental income influence students' academic performance?
3. In what ways does the educational level of parents affect students' grades?

In answering the following questions, the analysis seeks to give a better understanding of how socioeconomic factors interact with academic achievement, providing useful information that can inform educational strategies.

Working methodology

The research approach taken in this study adopts a systematic way of investigating the correlation between the academic performance of students and the factors affecting it, including playing hours, parents' income, and education levels of the parents. The procedure can be divided into a number of major steps to guarantee a comprehensive and precise investigation of the data.

- **Data Gathering**

The data utilized for this analysis was sourced from Kaggle Datasets, a well-known website that offers a number of publicly accessible datasets for research and data science. The dataset contains information related to students' grades, recreational activities, family income, and parents' education levels. This data was downloaded and loaded into a pandas DataFrame for processing and analysis.

- **Data Cleaning and Preprocessing**

The data cleaning phase is a fundamental step in any analytical project, as it involves examining missing values, duplicates, and ensuring the appropriateness of data types used.

In this project, we found missing values in the "Father Education" column. We addressed this issue by replacing the missing values with the mean, aiming to reduce their negative impact on the analysis. As for the duplicate values, which appeared as completely repeated rows, they were removed to avoid inflating the results and to achieve higher accuracy.

```
df.head()
```

	Sex	School Code	Playing Years	Playing Often	Playing Hours	Playing Games	Parent Revenue	Father Education	Mother Education	Grade	percentage
0	0	1	1	2	1	1	4	NaN	5	77.50	7750,00%
1	1	1	1	3	1	1	1	NaN	3	83.00	8300,00%
2	0	1	0	0	0	0	1	3.0	3	80.00	8000,00%
3	0	1	3	5	1	1	2	2.0	3	45.00	4500,00%
4	1	1	1	1	2	1	1	3.0	4	85.00	8500,00%

Working methodology

Additionally, we verified the data types and found that some columns were being read in inappropriate formats. For example, the percentage column was converted to a decimal type for use in the graph.

The data types for each column were correctly established (e.g., changing "Grade" to a numeric type) to facilitate proper calculations and comparisons. In addition, several columns in the data used ordinal numeric scales of 1 to 5 to represent qualitative attributes. For example, the variables "Playing Years," "Playing Often," "Playing Hours," "Parent Revenue," "Father Education," and "Mother Education" represent ordered experience, frequency, socioeconomic status, or education levels. These coded values were carefully interpreted in the analysis, with each numeric label having an equivalent meaningful real-world category (e.g., values of "Playing Often" indicate the number of days in a week that a student typically plays). Gender was encoded as 0 for female and 1 for male.

```
[ ] df["Father Education"]=df['Father Education'].fillna(df['Father Education'].mean())
    df.isnull().sum()
```

	0
Sex	0
School Code	0
Playing Years	0
Playing Often	0
Playing Hours	0
Playing Games	0
Parent Revenue	0
Father Education	0
Mother Education	0
Grade	0
percentage	0

dtype: int64

Working methodology

- **Exploratory Data Analysis (EDA)**

Having cleaned and preprocessed the data, we conducted an exploratory data analysis (EDA) to get a sense of the underlying structure and distributions of the dataset. From descriptive statistics such as the mean, median, and standard deviation, we gained information about the central tendency and variability of key variables such as grades, hours played, and parents' income.

We also performed some visualization to illustrate the relationships among variables, even though this step was kept to a minimum in this approach. EDA served as the foundation for identifying potential relationships that we would examine more closely in later steps.

- **Hypothesis Formulation and Testing**

From the preliminary results of EDA, we developed a list of hypotheses for relationships among the major variables. Some of the questions we wanted to address were:

1. Does increased time playing translate to worse grades?
2. Is there a connection between increased parental income and increased academic performance?
3. In what ways does parental education level influence student grades?
4. To test these hypotheses, we sorted the data based on specific criteria (e.g., students who played over 3 hours) and calculated mean grades for each subset.

- **Statistical Analysis**

In order to evaluate the intensity of the variable relationships, we conducted simple statistical computations such as contrasting mean grades against various income levels, parental education groups, and hours played.
Simple arithmetic and filtering were also applied in examining the contribution of varying factors to academic performance.

Conclusion and Insights Lastly, once the analysis was done, we consolidated the findings to draw actionable insights.
We aimed to address the hypotheses laid out at the onset and give an understanding of the impact of family background and individual habits on academic performance.
These insights were then distilled into a brief to identify key takeaways for educators, policymakers, and researchers.

Show Results

1. Parental Income and Academic Performance

One of the main areas we explored was the relationship between parental income and student grades.

We examined the link between parental income and student academic performance. The analysis revealed that 184 students who scored grades above 70 came from families with income higher than 3. suggesting a positive relationship. In contrast, 4 students with below 70 had a family income below 3, These results highlight the potential impact of socioeconomic status on student achievement and the importance of supporting low-income students.

	Condition	Number of Students
0	Grade > 70 and Income > 3	184
1	Grade < 70 and Income < 3	4
2	Grade > 70 and Income <= 3	41

2. Playing Hours and Grades

In this analysis, we focused on exploring the relationship between Playing Hours and Grades. The main goal was to understand if the time students spend playing affects their academic performance. We divided the students into two categories: those who spent more than 3 hours per day playing and those who spent less than 3 hours per day playing.
Initial Findings:

- For the students who spent more than 3 hours playing, the average grade was calculated to be 72.5. Meanwhile, students who played less than 3 hours had an average grade of 77.
- Interestingly, we found that students who played more than 3 hours per day achieved lower grades on average. However, some students in this group still had grades above 80, indicating that the relationship between playing and grades is not always straightforward.
-

In conclusion, while playing hours can affect academic performance, the impact varies depending on other factors like study habits, time management, and individual motivation.

Show Results

3. Mother's Education and Student Performance

We examined the relationship between mothers' education level and student grades. The analysis showed that 563 students either had mothers with education levels above 3 or achieved grades above 70, indicating a positive association. Meanwhile, 341 students with mothers' education at or below 3 generally had lower academic performance.

	Condition	Number of Students
0	Grade < 70 and Mother Education < 3	342
1	Grade > 70 and Mother Educatione > 3	578

The results indicate that most students with mothers who have higher education levels tend to achieve better grades, confirming the positive impact of maternal education on academic performance.

4. Playing Hours and Grade Comparison

This analysis further explores the impact of playing hours on grades by comparing students who spent more time playing to their academic performance.
Key Findings:

- For students who spent more than 3 hours playing, 18% of them scored above 80 on their grades, while 32%scored below 60. This suggests that, although some students spend more time playing and still get high grades, there is a notable proportion whose performance drops.
- On the other hand, students who played less than 3 hours showed a stronger performance, with 40% of them scoring above 80, and only 10% scoring below 60.
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In conclusion, while there is some evidence that students who spend more time playing may have lower grades, this relationship is not strong enough to draw definitive conclusions. It suggests that time management and prioritizing study are critical for students who want to excel academically while still engaging in recreational activities.

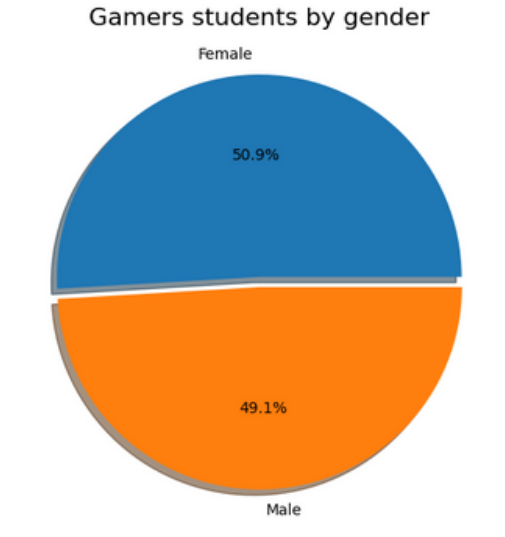
Show Results

5. Gamers students by gender

This pie chart shows the percentage of male and female students who play games.
We used a pie chart because it's a simple and visual way to represent proportions.

- Each slice of the chart represents a gender.
- The percentage is written inside each slice.

From this chart, we can easily see whether males or females make up the larger group of gamer students.
This helps us better understand the general trends in student gaming behavior.

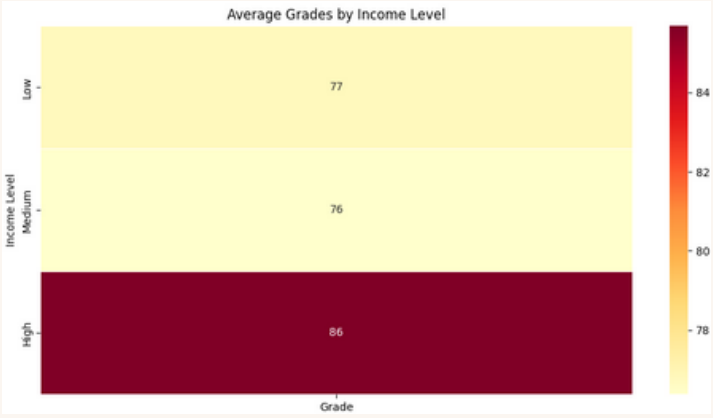


6. Average Grades by Income Level

This heatmap displays the average grades of students based on their parents' income level.

The income was categorized using a numerical scale ranging from 1 to 3, where:

- 1 or less represents Low income,
- More than 1 up to 3 represents Medium income,
- Above 3 represents High income.



For each income level, we calculated the average student grades and visualized the results using a heatmap:

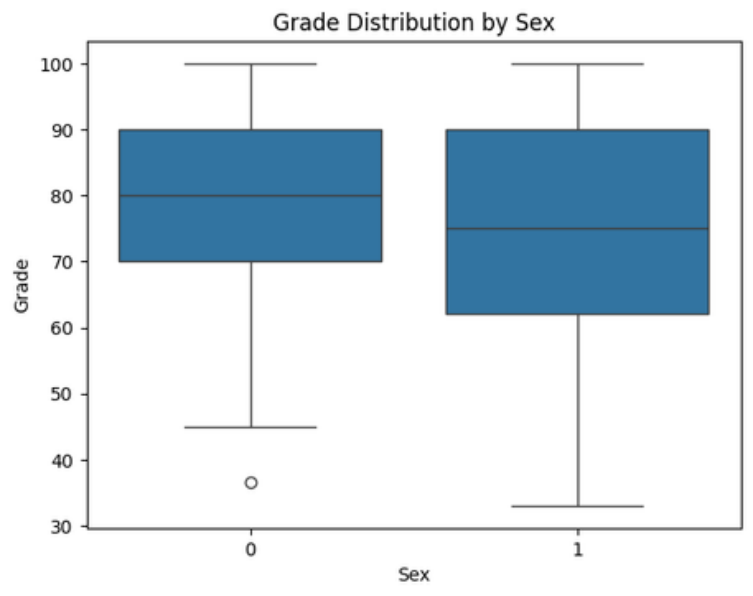
- Each colored box represents one income category,
- The darker the color, the higher the average grade,
- The numbers inside each box show the exact average.

This chart helps us quickly identify whether there's a relationship between income level and academic performance.

Show Results

7. Grade Distribution by Sex

Based on the boxplot, we can see that the average grades of females are higher than those of males.
Female grades are mostly distributed between 70 and 90, while male grades range approximately from 62 to 90.
Male grades show more variation, meaning their performance is more spread out, while female grades are more consistent and closer together.
We also notice a low outlier among the females, while no outliers appear for the males in this plot.

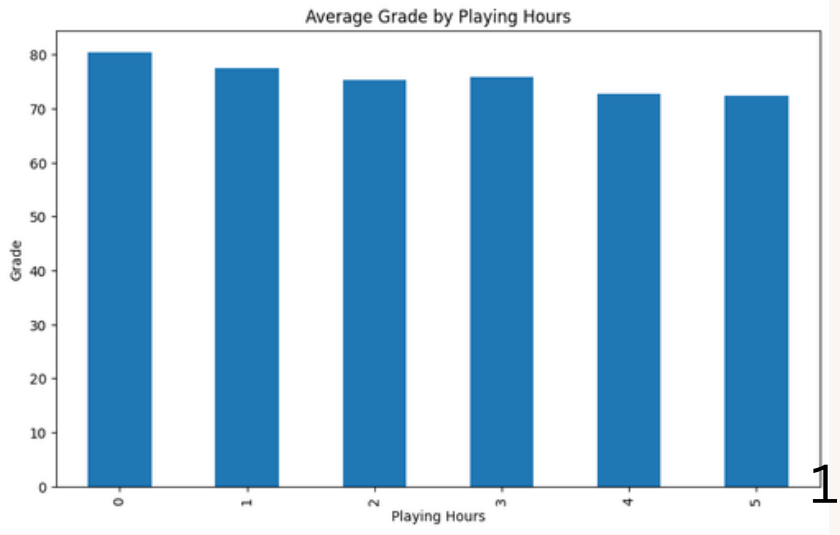


8. Average Grade by Playing Hours

Based on the bar chart, we can see that students who don't play at all (0 hours) have the highest average grades, around 80.

As the number of playing hours increases, the average grades decrease.

For example, students who play for 2 hours have an average grade of about 75, while those who play for 5 hours have an average grade of about 70.



Conclusions and Recommendations

Conclusions

The analysis shows that students' academic performance is influenced by several key factors. Higher parental income and a higher level of maternal education are associated with better grades. Students who spend more time playing tend to have lower academic achievement, although some exceptions exist. Additionally, female students generally perform better and show more consistent results compared to male students.

Recommendations

Based on these findings, it is recommended to support students from low-income families through academic and financial programs. Schools should also encourage parental involvement, especially by raising awareness among mothers about their role in student success. Time management training can help students balance study and leisure effectively. Furthermore, targeted support programs can help improve academic outcomes for male students.

References.

- https://www.kaggle.com/datasets/willianoliveiragibin/games-and-students